

SuViet360 - Interactive Vietnamese History Learning Platform Integrating a 2D RPG Game and Audio Podcast

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Context — Problem & Solution (Concise)

Problem

- History feels dry and hard to retain for learners.
- Traditional teaching lacks interactive visual tools.
- Students lack reliable, searchable history resources.
- Online historical content is fragmented and inaccurate.
- No engaging public forum for youth history exchange.

Solution

- SuViet360 combines AI and gamified history learning.
- RAG AI assistant offers accurate answers 24/7.
- Learn via minigames, audio-first podcasts and lessons.
- XP system and leaderboards encourage healthy competition.
- Community connection across web and mobile platforms.



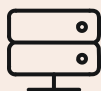
System Architecture & Technology Stack

Layered architecture designed for scalability, responsiveness and rich AI features.



Frontend

Next.js 16 (Web), React Native Expo (Mobile), Phaser.js for minigames, Redux Toolkit.



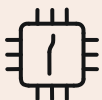
Backend

Node.js + Express.js with Socket.IO for real-time chat and APIs.



Data & Cache

MongoDB Atlas for content & users; Redis Cloud for caching and session store.



AI / GPU

RunPod serverless GPU running Qwen3-0.6B + LoRA; FAISS vector DB for RAG retrieval.



Integrations

Sepay webhook (payments), Resend (email), Cloudinary (media hosting).

Actors & Core Features

Students / Users

AI history Q&A, lessons
& podcasts, minigames,
real-time chat, VIP
subscriptions.

Teachers / Contributors

Audit historical
accuracy, test play
minigame, approve or
reject podcast or game
content with revision
feedback

Staff / Content Team

Upload podcast, process
game request, moderate
community blog & handle
content report

Admins

Manage user
roles(Student/Staff/
Teacher/Admin), monitor
platform analytics &
track VIP subscription

Main Flow 1 — RAG Chatbot (AI Q&A)

01

Step 1

User submits a history question via the AI chatbot UI.

02

Step 2

REST API forwards request to RunPod serverless GPU instance.

03

Step 3

FAISS vector search retrieves relevant Vietnamese sources (RAG).

04

Step 4

Qwen3 + LoRA synthesize a concise, referenced answer.

05

Step 5

Response displays with references and confidence indicators.

Main Flow 2 — Interactive Learning & Gamification



1. Select a lesson from the curated learning path.
2. Engage with text, podcast audio or Phaser.js minigame.
3. Complete a short multiple-choice quiz after the lesson.
4. System awards XP, badges and writes progress to MongoDB.
5. Ranks update on national leaderboard, motivating learners.

☐ XP and badges designed to promote retention and repeat engagement.

Main Flow 3 — VIP Subscription & Sepay Payments

1

Step 1

User chooses a VIP plan (monthly / yearly) on Subscription page.

2

Step 2

System generates a unique transaction code and displays VietQR.

3

Step 3

User scans VietQR with banking app and completes transfer.

4

Step 4

Sepay posts HTTPS webhook to backend confirming payment.

5

Step 5

Backend activates VIP access and send real-time notification.

Main Flow 4 — Content Management & Moderation



Step 1

Staff submits a new history lesson content (status: Pending).

Step 2

Teacher receives notification and audits historical accuracy

Step 3

Teacher approves or rejects (with feedback) via moderation UI.

Step 4

Approved posts publish; rejected posts return to staff for edits.

Step 5

Redis cache invalidated so updates appear instantly to all users.

Main Flow 5 — VIP Lesson Request

01

Step 1

VIP user submits a custom lesson request (title, historical period, description). Status: Pending.

02

Step 2

Teacher receives request, reviews topic, and accepts with pedagogical notes. Status: Accepted.

03

Step 3

Teacher starts working on content (creates podcast / requests minigame). Status: InProgress.

04

Step 4

Teacher completes content and links the result podcast to request. Status: Completed.

05

Step 5

Student receives real-time notification and accesses personalized lesson.

- 📋 Premium VIP users can request custom history lessons on specific topics. Teachers accept, create tailored podcast content, and deliver personalized learning experiences. Status flow: Pending → Accepted → InProgress → Completed.



Achievements, Constraints & Next Steps — Q&A

Achievements

AI RAG in Vietnamese on RunPod; unified Web & Mobile; Sepay payments automated.

Limitations

Cold-start latency for serverless GPU can be 30-40s on first request.

Next Directions

AI-generated exam creation, dynamic lesson system architecture.

Thank You / Q&A

Thank you to instructors and reviewers – we welcome questions and feedback.

📧 Contact: project lead luumynhathuy@gmail.com • Demo access available upon request.